Writing Correct Formula

Complete the following table by writing the correct formula for each combination of ions.

	F^-	Cl^-	O^{2-}	S^{2-}	N^{3-}
Na^+					
Li^+					
Ag^+					
K^+					
Cu^+					
Mg^{2+}					
Ba^{2+}					
Fe^{2+}					
Cu^{2+}					
Al^{3+}					
Cr^{3+}					
Fe^{3+}					
Sn^{4+}					
Pb^{4+}					

Note: Some metals can have more than one valency. This is indicated in the name by using roman numerals to state the valency.

i.e. iron (III) oxide Fe^{3+} combines with O^{2-} to give Fe_2O_3
tin (IV) sulfide Sn^{4+} combines with S^{2-} to give SnS_2

	bromide	iodide	oxide	sulfide	nitride
potassium					
copper (I)					
calcium					
iron (III)					
cobalt					
chromium (III)					
tin (II)					

Remember:

Do not change the formula of the polyatomic ion.

Brackets are only used when more than one polyatomic ion is needed.

	NO_3^-	OH^-	CH_3COO^-	SO_4^{2-}	CO_3^{2-}	PO_4^{3-}
Na^+						
Cu^+						
Cu^{2+}						
Fe^{2+}						
Fe^{3+}						
Sn^{4+}						
Zn^{2+}						
Al^{3+}						
NH_4^+						

Write the correct formula for each of the following compounds.

copper (II) chloride		ammonium carbonate	
aluminium oxide		magnesium iodide	
iron (II) phosphate		aluminium nitride	
gold (I) sulfide		barium ethanoate	
potassium sulfate		potassium hydrogencarbonate	
copper (I) carbonate		copper (II) sulfite	
calcium hydroxide		lead (II) nitride	
tin (II) chloride		tin (IV) nitride	
chromium (III) oxide		cobalt sulfate	
lead (IV) oxide		copper (II) hydroxide	
ammonium chloride		zinc phosphate	
iron (III) chloride		ammonium hydrogensulfate	

Name the following ionic compounds.

a) $NaCl$ _____

b) NH_4NO_3 _____

c) Ag_2O _____

d) $FeSO_4$ _____

e) $Fe_2(SO_4)_3$ _____

f) SnO_2 _____

g) $CuCO_3$ _____

h) K_3PO_4 _____

i) PbO_2 _____

j) Zn_3N_2 _____